Fakulteta za informacijske študije
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lab07

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```
~/Desktop
cat install.sh
echo "Installing demo software ..."
mkdir src
touch run.sh
echo "echo Hello" > run.sh
mv run.sh src
sh src/run.sh%
~/Desktop
```

```
~/Desktop
sha256sum install.sh > install.sh.sha256

~/Desktop
cat install.sh.sha256
e277935b53a8b2bec27536e89fdfd5e72e67b6b5034ce3d985fef89c70d438e3  install.sh
~/Desktop
```

```
~/Desktop                                                    38s
gpg --full-generate-key
gpg (GnuPG) 2.4.4; Copyright (C) 2024 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:
   (1) RSA and RSA
   (2) DSA and Elgamal
   (3) DSA (sign only)
   (4) RSA (sign only)
   (9) ECC (sign and encrypt) *default*
  (10) ECC (sign only)
```

```
~/Desktop                                                          ✔ | 5s
ls
Dockerfile       install.sh              secret.txt       secrets.env.gpg
bob_pubkey.asc   install.sh.sha256       secret.txt.gpg   webgoat-sbom.json
dvja-sbom.json   install.sh.sha256.sig   secrets.env
~/Desktop                                                          ✔
```

```
~/Desktop
echo "echo MALICIOUS CODE EXECUTED" >> install.sh
~/Desktop
sha256sum -c install.sh.sha256
install.sh: FAILED
sha256sum: WARNING: 1 computed checksum did NOT match
~/Desktop
```

```
gpg --verify install.sh.sha256.sig install.sh.sha256
gpg: Signature made Mon Apr 27 18:37:20 2026 CEST
gpg:                using RSA key 640D0FE2C9EBB1D048252EEABEC972E47E3C105B
gpg: Good signature from "peer@example.com" [ultimate]
~/Desktop                                                          ✔
echo "echo MALICIOUS CODE EXECUTED" >> install.sh
~/Desktop                                                          ✔
sha256sum -c install.sh.sha256
install.sh: FAILED
sha256sum: WARNING: 1 computed checksum did NOT match
~/Desktop                                                          1 x
```

Why should you never execute scripts without inspection?

You should never execute scripts without inspection because they may contain malicious code that can harm the system, steal data, or install malware.

What does SHA256 guarantee?

SHA256 guarantees file integrity. It ensures that the file has not been modified.

What does GPG guarantee?

GPG guarantees authenticity and origin of the file. It proves who created or signed the file.

Can SHA256 prove the author?

No, SHA256 cannot prove the author. It only verifies that the file has not changed.

Why is signing the checksum important?

Signing the checksum ensures that the checksum itself is authentic and not modified by an attacker.

What happened after modification?

After modifying the script, the checksum verification failed, indicating that the file integrity was compromised.